



STROKE QUALITY IMPROVEMENT PROGRAM

Document ID: SU-009-PLAN-01 | Parent Policy: SU-009 — Stroke Performance Improvement, Data Registry and Peer Review | Version: 1.0 | Controlled Copy

Certification Alignment Note: All example 'Current' performance values and illustrative case counts must be replaced with verified KFCH registry/quality data before approval or submission as accreditation evidence.

Supporting Documentation for AHA/ASA Stroke Center Certification

Documents Included in This Set

- 1. Stroke Quality Improvement Plan 2026
- 2. Stroke Program Complication Monitoring Plan
- 3. Stroke Procedural Quality Dashboard
- 4. Patient-Centered Stroke Research
- 5. Aligning Quality Improvement with IPC Meeting Minutes

This document set consolidates the Stroke Program's quality improvement plan, complication and procedural monitoring, patient-centered research documentation, and guidance for reflecting quality activity in Stroke Interprofessional Committee (IPC) meeting minutes. Together, these sections demonstrate the continuous cycle of measurement → review → improvement → education → reassessment expected for AHA Stroke Certification.

DOCUMENT 1

Stroke Quality Improvement Plan 2026

Purpose

To continuously improve the quality, safety, and timeliness of stroke care using performance metrics, education, and multidisciplinary review.

Target Time Goals

Measure	Current	Target	Review Frequency
Door-to-Stroke Alert	8 min	≤ 10 min	Monthly
Door-to-CT Start	24 min	≤ 20 min	Monthly
Door-to-CT Interpretation	48 min	≤ 45 min	Monthly
Door-to-Tenecteplase/Alteplase	71 min	≤ 60 min	Monthly
Door-to-CTA	35 min	≤ 30 min	Monthly
Door-to-Telestroke Consult	38 min	≤ 20 min	Monthly
Door-to-Groin Puncture (LVO)	110 min	≤ 90 min	Quarterly
Dysphagia Screening	82%	100%	Monthly
NIHSS Documentation	90%	100%	Monthly

Quality Improvement Action Plan

Issue Identified	Root Cause	Improvement Action	Responsible	Timeline
Door-to-Needle > 60 min	Delayed CT notification	Stroke Alert overhead paging	ED Director	July
Delayed Telestroke	Neurologist activation	Single-call activation process	Stroke Coordinator	August
Missing NIHSS	Staff competency	NIHSS refresher training	Education Dept	Monthly
Delayed Dysphagia Screen	Nursing awareness	Bedside competency	Nurse Educator	Monthly

Staff Education Plan

- Monthly Stroke Education
- Mock Stroke Code every month
- NIHSS competency twice/year
- Dysphagia screening competency
- EMS feedback meetings

- Radiology education
- Laboratory turnaround education

Communication Plan

Demonstrates communication with frontline staff, for example:

- Monthly Stroke Dashboard emailed
- Quality Board in ED
- Stroke Newsletter
- Staff huddles
- Department meetings
- Stroke Champion meetings

Evidence to Upload

- Stroke Dashboard
- Education attendance sheets
- Mock Code reports
- Email communications
- IPC meeting minutes discussing KPIs

DOCUMENT 2

Stroke Program Complication Monitoring Plan

Complications Monitored

Complication	Target
Symptomatic Intracranial Hemorrhage after thrombolysis	< 6%
Aspiration Pneumonia	< 10%
DVT	< 5%
Pulmonary Embolism	0%
Falls	< 2%
Pressure Injury	< 3%
UTI	< 5%
Seizure after Stroke	Monitor
Cerebral Edema requiring surgery	Monitor
Mortality	Monitor monthly
Readmission within 30 days	Monitor

QI Initiative Example

Illustrative Case

Problem: Aspiration pneumonia increased to 9%.

Actions:

- Mandatory swallow screen
- Speech Therapy within 24 hours
- Nursing education
- Audit compliance

Outcome: **Reduced to 4%.**

DOCUMENT 3

Stroke Procedural Quality Dashboard

Accreditation Note

- AHA requires evidence that procedures are reviewed by the Stroke Interprofessional Committee (IPC).

Procedural Complication Summary (De-identified Example)

Procedure	Cases	Complications
IV Thrombolysis	48	sICH: 2
Mechanical Thrombectomy	22	Vessel perforation: 0
Carotid Endarterectomy	10	Cranial nerve injury: 0
Carotid Stenting	8	Access hematoma: 1
Diagnostic Angiography	36	Contrast reaction: 1

IPC Review — Example Meeting Discussion

Agenda: Procedural Complication Review

Discussion:

- One access-site hematoma after carotid stenting
- Root cause reviewed
- Ultrasound-guided femoral puncture recommended
- Updated checklist approved
- Education completed for interventional team

Additional Graphs to Include

- Monthly complication trend
- Quarterly trend
- Benchmark comparison

DOCUMENT 4

Patient-Centered Stroke Research**Note**

- This accreditation requirement is commonly misunderstood. Research may take any of the three forms below.

Option 1 (Preferred) — Registry / Trial Participation

Hospital participates in:

- Get With The Guidelines–Stroke Registry
- Saudi Stroke Registry
- Ministry of Health Stroke Registry
- Clinical trials

Provide:

- Participation letter
- Registry screenshots
- Research committee approval

Option 2 — Stroke Quality Improvement Project**Example: Reducing Door-to-Needle Time Using Lean Methodology**

Objective: Reduce DTN from 72 minutes to 55 minutes.

Methods:

- Stroke Alert redesign
- CT prioritization
- Monthly education
- Mock stroke codes

Results: DTN improved from 72 min → 58 min.

Conclusion: Improved patient outcomes and faster thrombolysis.

Option 3 — Patient Education Research

Examples:

- Patient satisfaction after stroke education
- Medication adherence
- Quality of life surveys
- Caregiver education effectiveness

Documents to Upload

- Research proposal
- IRB approval (if applicable)
- Stroke registry participation
- Poster presentation
- Abstract
- Quality Improvement project report
- Published paper (if available)

DOCUMENT 5

Aligning Quality Improvement with IPC Meeting Minutes

Stroke Interprofessional Committee (IPC) meeting minutes should clearly reflect discussion of the following items:

- Review of door-to-needle, door-to-CT, and door-to-consult performance against targets.
- Analysis of stroke complications (e.g., symptomatic intracranial hemorrhage, aspiration pneumonia, falls) with root causes and corrective actions.
- Review of periprocedural outcomes for thrombolysis, thrombectomy, carotid interventions, and angiography, including any adverse events and improvement measures.
- Updates on staff education (NIHSS competency, mock stroke codes, dysphagia screening, EMS feedback).
- Progress of ongoing quality improvement projects and patient-centered research or registry participation, with action items and responsible persons.

Core Expectation

- Together, these documents demonstrate a continuous cycle of measurement → review → improvement → education → reassessment, which is the core expectation of AHA Stroke Certification.